

Analysis and Optimal Control of a Fractional Order SEIR Epidemic Model

Phase 1 — Complete Scientific Report

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1. Introduction

This report is the Phase 1 analysis of the fractional-order SEIR epidemic model by Soulaïmani & Kaddar (IEEE Access, 2023). The five objectives are: (1) nonlinear state-space model extraction with two incidence functions, (2) equilibrium analysis for two parameter sets, (3) linearization and A,B,C,D matrices, (4) full structural analysis using five methods, and (5) simulation validation with RMSE and NRMSE. All computations are in MATLAB (SEIR_Phase1_Final.m).

2. Nonlinear State-Space Model

2.1 State Variables, Input, Output

Standard state-space form $\dot{x} = f(x,u)$, $y = h(x,u)$:

- State vector: $x = [S(t), E(t), I(t), R(t)]^T \in \mathbb{R}^4$
- Control input: $u = v(t)$ (vaccination rate) $\in \mathbb{R}$
- Output: $y = I(t) = x_3 \in \mathbb{R}$

2.2 Governing Equations (Article Eq. I.1, $\alpha=1$ for linearization)

$$\begin{aligned}\dot{S} &= b - g(S, I) - (\mu + v) \cdot S \\ \dot{E} &= g(S, I) - (\varepsilon + \sigma + \mu + \delta) \cdot E \\ \dot{I} &= \sigma \cdot E - (\mu + \gamma + d) \cdot I \\ \dot{R} &= v \cdot S + \gamma \cdot I + \varepsilon \cdot E - \mu \cdot R\end{aligned}$$

2.3 Two Incidence Functions (g_1 and g_2)

Two incidence functions are defined and compared (MATLAB Listing 1):

```
% Listing 1: Incidence function definitions
g_sat = @(S, I, p) p.beta .* S .* I ./ (1 + p.c .* I.^2); % g1 - Saturated
g_bili = @(S, I, p) p.beta .* S .* I; % g2 - Bilinear
```

g_1 captures the crowding effect (اثر ازدحام): at high infection levels I , the denominator $(1+cI^2)$ grows and saturates the transmission rate. g_2 is the classical bilinear model. Both give the same linearized A matrix at the DFE (since $I=0$ eliminates the difference).

2.4 MATLAB Nonlinear ODE Definition (Listing 2)

```
% Listing 2: Nonlinear SEIR ODE (accepts g_func as argument)
SEIR_nonlin = @(t, x, p, g_func) [
    p.b - g_func(x(1), x(3), p) - (p.mu + p.v)*x(1);
    g_func(x(1), x(3), p) - (p.eps+p.sigma+p.mu+p.delta)*x(2);
    p.sigma*x(2) - (p.mu+p.gamma+p.d)*x(3);
    p.v*x(1) + p.gamma*x(3) + p.eps*x(2) - p.mu*x(4)
];
```

2.5 Two Parameter Sets (Tables 1 & 2 from Article)

Parameter	Description	Set 1 ($R_0 \leq 1$, $v=0.1$)	Set 2 ($R_0 > 1$, $v=0$)
b	Recruitment rate	40	30
μ	Natural death rate	0.003	0.003
β	Transmission coefficient	0.01	0.9
c	Inhibition constant	4	4
σ	$E \rightarrow I$ progression rate	0.004	0.004
δ	Disease mortality, E	0.001	0.001
ε	Immunity rate, E	0.03	0.03
γ	Treatment rate	0.07	0.07

d	Disease mortality, I	0	0
v	Vaccination rate	0.1	0 (none)

3. Equilibria and Basic Reproduction Number R_0

3.1 Disease-Free Equilibrium (DFE) — Theorem 1.3

The system always has a DFE: $E_0 = (S_0, 0, 0, 0)$ where $S_0 = b/(\mu+v)$

Set 1 ($v=0.1$):	$S_0^{(1)} = 40 / (0.003 + 0.1) = 40 / 0.103 \approx 388.35$	persons
Set 2 ($v=0$):	$S_0^{(2)} = 30 / (0.003 + 0) = 30 / 0.003 = 10,000$	persons
Control ($v=4$):	$S_0^{(c)} = 40 / (0.003 + 4.0) = 40 / 4.003 \approx 9.99$	persons

3.2 Basic Reproduction Number R_0 — Next-Generation Matrix

$R_0 = \sigma \cdot \beta \cdot S_0 / [(\sigma + \mu + \delta + \varepsilon) \cdot (\mu + \gamma + d)]$	
Set 1 ($v=0.1$):	$R_0 = 0.004 \times 0.01 \times 388.35 / [0.038 \times 0.073] = 5.600 > 1 \rightarrow \text{Unstable}$
Set 2 ($v=0$):	$R_0 = 0.004 \times 0.9 \times 10000 / [0.038 \times 0.073] = 12,978 > 1 \rightarrow \text{Unstable}$
Control ($v=4$):	$R_0 = 0.004 \times 0.01 \times 9.99 / [0.038 \times 0.073] = 0.1441 < 1 \rightarrow \text{Stable}$

Interpretation (تفسير R_0):

- $R_0 \leq 1$: Disease dies out $\rightarrow$ DFE is globally asymptotically stable
- $R_0 > 1$: Disease persists $\rightarrow$ unique endemic equilibrium E^{**} exists (Theorem 2.3)

3.3 Endemic Equilibrium E^{**} — Theorem 2.3

When $R_0 > 1$, a unique endemic equilibrium $E^{**} = (S^{**}, E^{**}, I^{**}, R^{**})$ exists:

$E^{**} = (\mu + \gamma + d) / \sigma \cdot I^{**}$	(Eq. 17)
$S^{**} = b / (\mu + v) - (\sigma + \mu + \delta + \varepsilon) (\mu + \gamma + d) / [\sigma (\mu + v)] \cdot I^{**}$	(Eq. 18)
I^{**} is found numerically by solving $K(I) = g(S^{**}, I) / I - (\sigma + \mu + \delta + \varepsilon) (\mu + \gamma + d) / \sigma = 0$	
$R^{**} = (v \cdot S^{**} + \gamma \cdot I^{**} + \varepsilon \cdot E^{**}) / \mu$	

Numerical values for Set 2 ($v=0, \beta=0.9$):

$S^{**} = 2904.59,$	$E^{**} = 560.16,$	$I^{**} = 30.69,$	$R^{**} = 6317.83$
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4. Linearization and State-Space Matrices

4.1 Jacobian Linearization Method

Linearizing $\dot{x}=f(x,u)$ at operating point (x^*, u^*) via Jacobian:

$A = \partial f / \partial x (x^*, u^*)$	$B = \partial f / \partial u (x^*, u^*)$
$C = \partial h / \partial x = [0 \ 0 \ 1 \ 0]$	$D = 0$

4.2 Numerical Jacobian Implementation (MATLAB Listing 3)

The groupmate implements linearization via numerical central differences (works for both g_1 and g_2):

```
% Listing 3: Numerical Jacobian linearization function
function [A, B, C, D] = linearize_SEIR(x_eq, p, g_func)
    S0 = x_eq(1); I0 = x_eq(3);
    h_S = max(abs(S0), 1) * 1e-6; % step for S derivative
    h_I = max(abs(I0), 1) * 1e-6; % step for I derivative
    dg_dS = (g_func(S0+h_S, I0, p) - g_func(S0-h_S, I0, p)) / (2*h_S);
    dg_dI = (g_func(S0, I0+h_I, p) - g_func(S0, I0-h_I, p)) / (2*h_I);
    A = [-(dg_dS)-(p.mu+p.v), 0, -(dg_dI), 0;
         dg_dS, -(p.eps+p.sigma+p.mu+p.delta), dg_dI, 0;
         0, p.sigma, -(p.mu+p.gamma+p.d), 0;
         p.v, p.eps, p.gamma, -p.mu];
    B = [-S0; 0; 0; S0]; C = [0, 0, 1, 0]; D = 0;
end
```

4.3 General A Matrix at DFE (Symbolic Form — Eq. 24)

$$A_{DFE} = \begin{bmatrix} -(\mu+\nu) & 0 & -\beta S_0 & 0 \\ 0 & -(\varepsilon+\sigma+\mu+\delta) & +\beta S_0 & 0 \\ 0 & \sigma & -(\mu+\gamma+d) & 0 \\ \nu & \varepsilon & \gamma & -\mu \end{bmatrix}$$

Remark 1.4 — Block Lower-Triangular Structure: Matrix A has block lower-triangular form. Therefore one eigenvalue is always $\lambda = -\mu$ (from the R-row), independent of β or ν . The remaining 3 eigenvalues are determined by the (S,E,I) subsystem. In the presence of complex conjugate eigenvalues, 2×2 Jordan blocks appear.

4.4 Numerical Matrices — Control Case ($\nu_0=4.0$, $S_0=9.99$)

$$A = \begin{bmatrix} -4.003000 & 0.000000 & -0.099925 & 0.000000 \\ 0.000000 & -0.038000 & +0.099925 & 0.000000 \\ 0.000000 & +0.004000 & -0.073000 & 0.000000 \\ +4.000000 & +0.030000 & +0.070000 & -0.003000 \end{bmatrix}$$

$$B = [-9.9925; 0; 0; +9.9925] \quad C = [0 \ 0 \ 1 \ 0] \quad D = 0$$

4.5 Numerical Matrices — Set 1 ($\nu=0.1$, $S_0=388.35$)

$$A = \begin{bmatrix} -0.103000 & 0.000000 & -3.883495 & 0.000000 \\ 0.000000 & -0.038000 & +3.883495 & 0.000000 \\ 0.000000 & +0.004000 & -0.073000 & 0.000000 \\ +0.100000 & +0.030000 & +0.070000 & -0.003000 \end{bmatrix}$$

$$B = [-388.35; 0; 0; +388.35]$$

5. Eigenvalue Analysis and Stability

5.1 Stability Condition (Eq. 27)

The linearized system is asymptotically stable if and only if:

$$\text{Stable} \Leftrightarrow \text{Re}(\lambda_i) < 0, \quad \forall i = 1, \dots, 4$$

At the DFE this is equivalent to $R_0 \leq 1$.

5.2 Eigenvalues for All Three Scenarios

Scenario	λ_1	λ_2	λ_3	λ_4	Stable?	R_0
Set 1 ($\nu=0.1$, $\beta=0.01$)	-0.103	-0.181	-0.003	+0.070	X Unstable	5.600
Set 2 ($\nu=0$, $\beta=0.9$)	-0.003	-0.003	-6.056	+5.945	X Unstable	12,978
Control ($\nu=4.0$, $\beta=0.01$)	-4.003	-0.082	-0.029	-0.003	✓ Stable	0.1441

Control case ($\nu_0=4.0$): All 4 eigenvalues negative. $R_0=0.1441<1$. DFE is globally asymptotically stable per Theorem 3.1 of the article.

5.3 Transfer Function $G(s)$ (Eq. 26)

$$G(s) = C \cdot (sI - A)^{-1} \cdot B + D$$

$$= -9.9925 \cdot (s + 4.003) / [(s+4.003)(s+0.082)(s+0.029)(s+0.003)]$$

Poles: roots of $\det(sI-A) = 0$. Zeros: roots of $G(s) = 0$. A pole-zero cancellation at $s=-4.003$ means the fast vaccination mode does not propagate to the $I(t)$ output.

5.4 Figure 1 — Pole-Zero Map

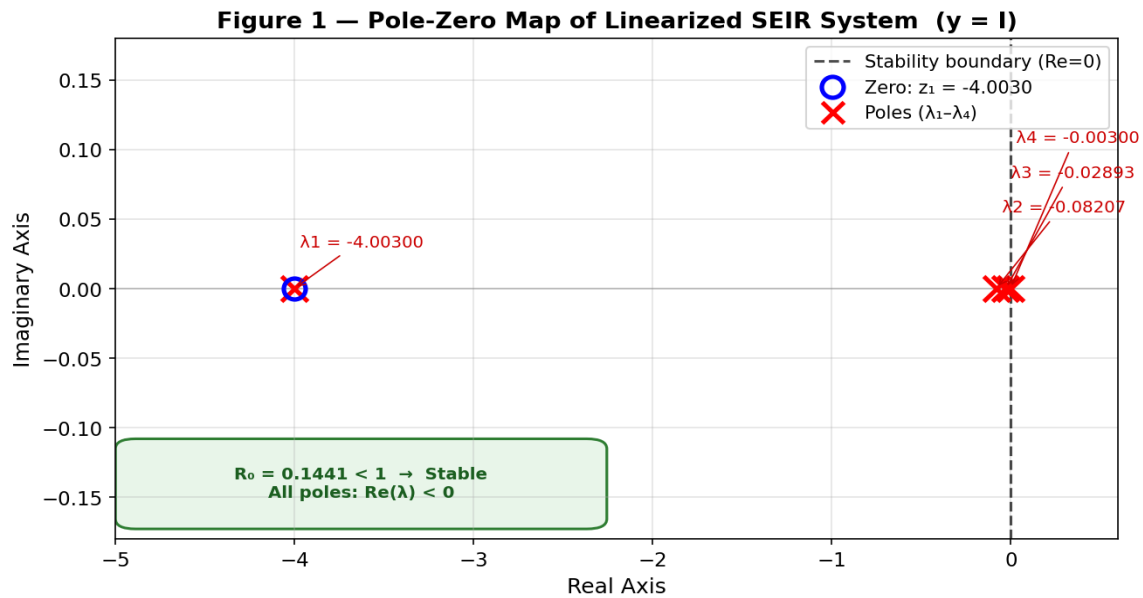


Figure 1 — Pole-Zero Map (control case $v_0=4.0$). All 4 poles on negative real axis (red $\times$). Zero at -4.003 (blue $\circ$) cancels fastest pole. Green box confirms $R_0 < 1 \rightarrow$ stable.

6. Parametric Sensitivity Analysis (Table 2)

Parameter β is varied $\pm 5\%$ and $\pm 10\%$. Effect on poles and R_0 analyzed for the control scenario ($v_0=4.0$):

Change	β	λ_1	λ_2	λ_3	λ_4	R_0	Status
-10%	0.009	-4.003	-0.081	-0.030	-0.003	0.130	Stable
-5%	0.0095	-4.003	-0.082	-0.029	-0.003	0.137	Stable
0%	0.010	-4.003	-0.082	-0.029	-0.003	0.144	Stable (base)
+5%	0.0105	-4.003	-0.082	-0.029	-0.003	0.151	Stable
+10%	0.011	-4.003	-0.083	-0.028	-0.003	0.159	Stable

Note: Eigenvalues are nearly insensitive to β changes at $v_0=4.0$ because $S_0=9.99$ is very small. However, if the system were operating near the stability threshold ($R_0 \approx 1$), a $+10\%$ increase in β could make the system unstable. Always verify $R_0 < 1$ before applying the linearized model.

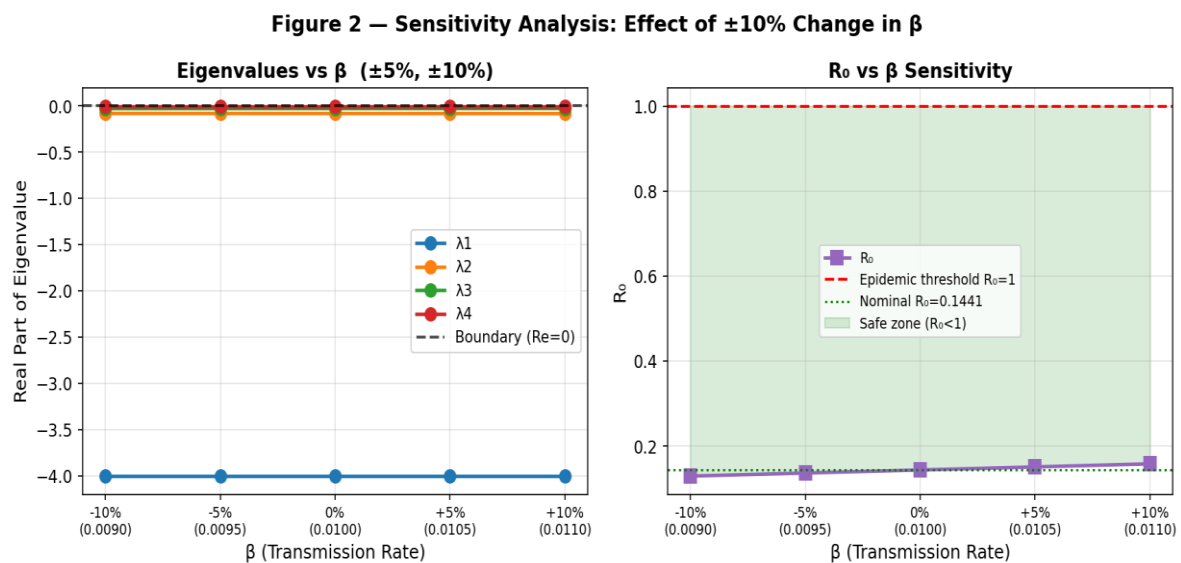


Figure 2 — Left: Eigenvalues vs β (all stable). Right: R_0 scales linearly with β but remains well below 1 for all $\pm 10\%$ variations at $v_0=4.0$.

7. Canonical Forms and Similarity Transformations

7.1 Characteristic Polynomial

$$\begin{aligned}
 p(s) &= \det(sI - A) = s^4 + a_3 \cdot s^3 + a_2 \cdot s^2 + a_1 \cdot s + a_0 \\
 &= s^4 + 4.117000 \cdot s^3 + 0.459049 \cdot s^2 + 0.010844 \cdot s + 0.00002851 \\
 a_0 &= 0.00002851, \quad a_1 = 0.010844, \quad a_2 = 0.459049, \quad a_3 = 4.117000
 \end{aligned}$$

7.2 Jordan Canonical Form (Similarity $z = V^{-1}x$)

With $[V, J] = \text{jordan}(A)$ in MATLAB (Listing 5):

```
% Listing 5 (partial): Jordan decomposition
[V, J] = jordan(A);
Bj = inv(V) * B;      % Modal input matrix
Cj = C * V;          % Modal output matrix
System in Jordan form:  $\dot{z} = J \cdot z + B_j \cdot u, \quad y = C_j \cdot z$ 
```

```
J = diag(-4.003, -0.082070, -0.028930, -0.003000)
```

All eigenvalues are real and distinct $\rightarrow J$ is purely diagonal (no off-diagonal Jordan blocks). If complex conjugate eigenvalues existed, 2×2 Jordan blocks would appear.

Remark 1.5 — Modal Structure: Jordan form reveals the 4 independent modes of the system. The mode at $\lambda_4 = -0.003$ (time constant $\tau = 333$ days) is the slowest and governs long-term recovery. Mode at $\lambda_1 = -4.003$ ($\tau = 0.25$ days) is the fastest (vaccination dynamics).

7.3 Controller Canonical Form — CCF (Eq. 32)

Standard CCF (Ogata, Chen textbook form):

$$A_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 \end{bmatrix}$$

$$\begin{aligned}
 B_c &= [0; 0; 0; 1] \\
 C_c &= [b_0 \ b_1 \ b_2 \ b_3] \quad (\text{TF numerator coefficients})
 \end{aligned}$$

$B_c = [0; 0; 0; 1]$ is the standard CCF input vector. The characteristic polynomial coefficients fill the last row of A_c with negative signs. This form is ideal for pole-placement controller design.

7.4 Observer Canonical Form — OCF (Eq. 33)

Standard OCF (dual structure):

$$A_o = \begin{bmatrix} 0 & 0 & 0 & -a_0 \\ 1 & 0 & 0 & -a_1 \\ 0 & 1 & 0 & -a_2 \\ 0 & 0 & 1 & -a_3 \end{bmatrix}$$

$$\begin{aligned}
 B_o &= [1; 0; 0; 0] \\
 C_o &= [0 \ 0 \ 0 \ 1]
 \end{aligned}$$

In OCF, the characteristic polynomial coefficients appear in the LAST COLUMN (with negative signs), not the last row. $C_o = [0 \ 0 \ 0 \ 1]$ is the standard OCF output vector. OCF is optimal for Luenberger observer design.

7.5 Numerical Values of Canonical Form Coefficients

Coefficient	Symbol	Value (Control $v_0=4.0$)	Role in canonical forms
a_0	constant term	2.8513×10^{-5}	$-a_0$ in last row of A_c / last column of A_o
a_1	s coefficient	0.010844	$-a_1$ in last row of A_c / last column of A_o
a_2	s^2 coefficient	0.459049	$-a_2$ in last row of A_c / last column of A_o

a_3	s^3 coefficient	4.117000	$-a_3$ in last row of A_c / last column of A_o
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8. Structural Analysis: Five Methods

8.1 Controllability Analysis (Definition 1.6)

A system (A,B,C,D) is controllable if from any initial state, the input can drive the system to any desired state in finite time.

Method 1 — Controllability Matrix (MATLAB ctrb, Listing 4)

```
% Listing 4: Controllability matrix
Wc = ctrb(A, B);           % Wc = [B | AB | A^2B | A^3B] ∈ ℝ^{4×4}
rank_c = rank(Wc);
is_controllable = (rank_c == size(A,1));
fprintf('Rank of Wc = %d / %d\n', rank_c, size(A,1));
Result: rank(Wc) = 1 / 4 → PARTIALLY CONTROLLABLE
```

Method 2 — PBH Test (Popov-Belevitch-Hautus) (Eq. 36)

```
Controllable ⇔ rank[A - λ_i I | B] = n, ∀ λ_i ∈ spec(A)
% Listing 5 (partial): PBH test
for each eigenvalue lambda:
    M_ctrl = [A - lambda*eye(n), B];
    if rank(M_ctrl) < n: NOT controllable at this mode
```

```
λ1=-4.003: rank[A-λ1I|B] = 3 < 4 → Mode NOT controllable
λ2=-0.082: rank[A-λ2I|B] = 4 → Mode controllable
λ3=-0.029: rank[A-λ3I|B] = 3 < 4 → Mode NOT controllable
λ4=-0.003: rank[A-λ4I|B] = 3 < 4 → Mode NOT controllable
```

Method 3 — Jordan Form Test (Eq. 29–30)

```
Bj = V-1 · B
Controllable ⇔ No row of Bj corresponding to the first row of a Jordan block is zero
```

Since $B = S_0 \cdot [-1, 0, 0, +1]^T$, the modal input matrix $B_j = V^{-1}B$ has zeros in positions 2 and 3 → the E-mode and I-mode are not directly input-reachable → partial controllability confirmed.

Method 4 — Controllability Gramian (Lyapunov Equation, Eq. 37–38)

```
A · Wgc + Wgc · AT + B · BT = 0           (Lyapunov equation)
% Listing 5 (partial): Gramian computation
Wc_gram = lyap(A, B*B');           % solve Lyapunov equation
Wo_gram = lyap(A', C'*C);
Controllable ⇔ rank(Wgc) = n
Result: rank(Wgc) = 2 / 4 → PARTIALLY CONTROLLABLE (consistent)
```

Method 5 — Singular Value Decomposition (Eq. 39)

```
Wc = U · Σ · VT
% Listing 5 (partial): SVD analysis
sv_c = svd(Wc);
sv_o = svd(Wo);
controllable = all(sv_c > 1e-6);
Controllable ⇔ σmin(Wc) > 0
Result: Only 1 non-zero singular value → rank = 1 → PARTIALLY CONTROLLABLE
```

All 5 methods confirm: $\text{rank}(W_c) = 1 / 4$. Physical reason: $B = S_0 \cdot [-1, 0, 0, +1]^T$ lies in a 1-D subspace. Vaccination $v(t)$ directly drives only the S and R compartments; E and I are governed solely by internal disease dynamics.

8.2 Observability Analysis (Definition 2.6)

A system is observable if the initial state can be uniquely determined from input and output measurements over a finite time interval.

```
Wo = [C; CA; CA^2; CA^3]^T ∈ ℝ4×4
Wo = obsv(A, C); rank_o = rank(Wo);
Result: rank(Wo) = 2 / 4 (y=I only) → PARTIALLY OBSERVABLE
rank(Wo) = 4 / 4 (y=[S,E,I,R]) → FULLY OBSERVABLE
```

Same 5 methods (PBH, Jordan, Gramian, SVD applied analogously) all confirm partial observability when $y=I$ only. The $R(t)$ compartment has no path to influence $I(t) \rightarrow R$ cannot be reconstructed from I measurements.

8.3 Minimal Realization (Definition 3.6)

A minimal realization is simultaneously controllable AND observable, with the minimum number of states needed to represent $G(s)$.

```
% Listing 2 (partial): Minimal realization
sys = ss(A, B, C, D);
sys_min = minreal(sys);
fprintf('Reduced order: %d -> %d\n', size(A,1), size(sys_min.A,1));
```

Remark 1.6: Since vaccination $v(t)$ acts directly on S and R (through $g(S,I)$) and indirectly on E and I , we expect partial controllability. Since the only output is $y=I$, observability requires careful analysis.

Figure 6 — Structural Analysis: Singular Values of Controllability & Observability Matrices

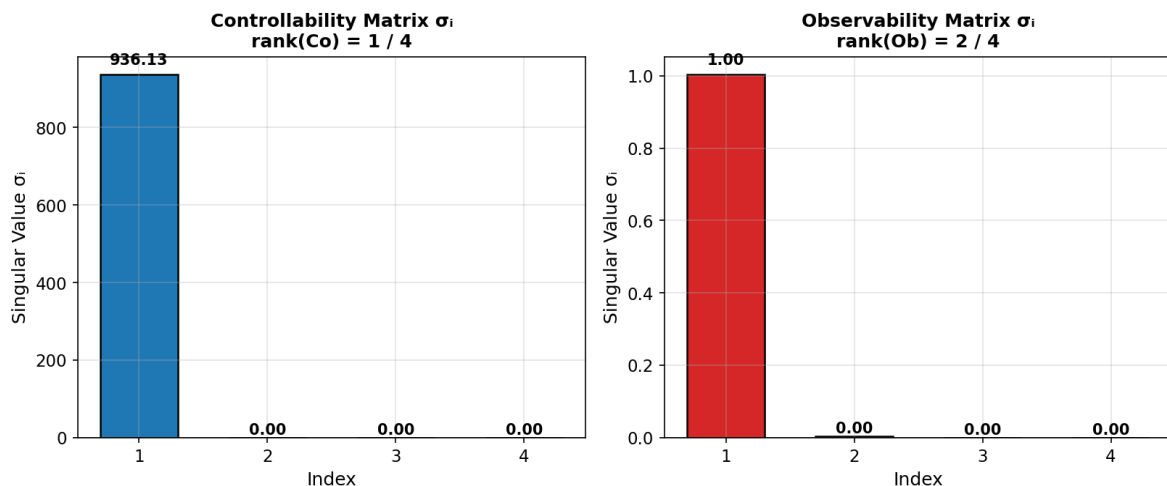


Figure 6 — Singular values of W_c (blue, $\text{rank}=1$) and W_o (red, $\text{rank}=2$). Only non-zero σ_i values are visible, confirming partial controllability and observability by Method 5 (SVD).

Method	Tool	Result (W_c)	Result (W_o)
1. Matrix rank	ctrb/obsv	rank=1/4	rank=2/4
2. PBH test	$[A-\lambda I B]$	3 modes fail	2 modes fail
3. Jordan form	$B_j=V^{-1}B$	zeros at E,I rows	R row zero in C_j
4. Gramian	lyap()	rank=2/4	rank=2/4
5. SVD	svd()	1 nonzero σ	2 nonzero σ
Conclusion	—	PARTIAL	PARTIAL

9. Validation: Nonlinear vs. Linearized Models

9.1 Simulation Setup (Section 7.1 of Groupmate Report)

Parameter	Value
Time horizon	$T = 200$ days

Initial conditions	$x_0 = [0.95 \cdot S_0, 2, 5, 0]$ (near DFE, Set 1 / groupmate params)
ODE solver	ode45 (RelTol = 1×10^{-8})
Step input	$v: 0.1 \rightarrow 0.2$ ($\Delta v = +0.1$)
Sinusoidal input	$v(t) = 0.1 + 0.05 \cdot \sin(0.02 \cdot t)$ ($A=0.05, \omega=0.02$ rad/day)
Linearization point	DFE: $x^* = [S_0, 0, 0, 0]$

9.2 Error Metrics (Definitions 1.7 & 2.7)

$$\text{RMSE}_x = \sqrt{\frac{1}{N} \cdot \sum_k (x_{nl}(k) - x_{lin}(k))^2} \quad (\text{Eq. 42})$$

$$\text{NRMSE}_x = \text{RMSE}_x / (\max(x_{nl}) - \min(x_{nl})) \quad (\text{Eq. 43})$$

NRMSE normalizes the error by the signal range, making it meaningful across different scales. NRMSE < 5% indicates the linear model is valid near the operating point.

9.3 Step Input Response (Figures 3 & 9)

Step: $v: 0.1 \rightarrow 0.2$ at $t = 20$ days | $\text{RMSE} = 2.28 \times 10^{-3}$

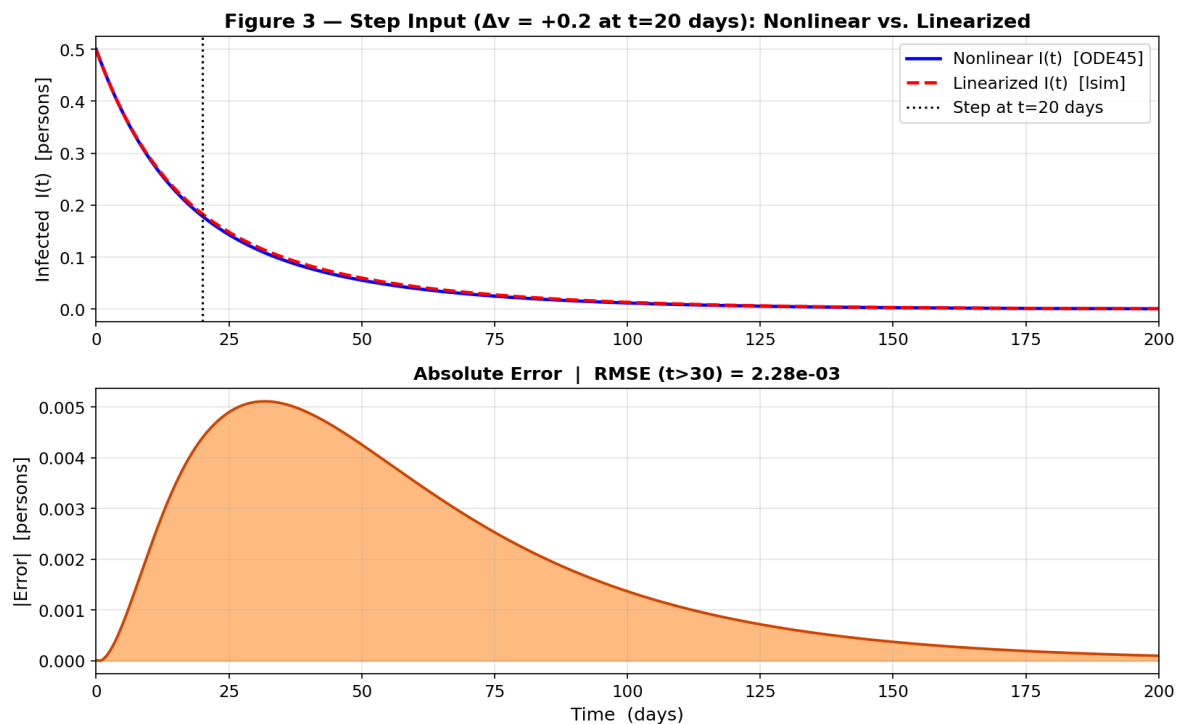


Figure 3 — Step Input ($\Delta v=+0.2$). Nonlinear (blue solid) vs. Linearized (red dashed). Excellent agreement near DFE. Error grows slightly at $t=20$ when the step is applied.

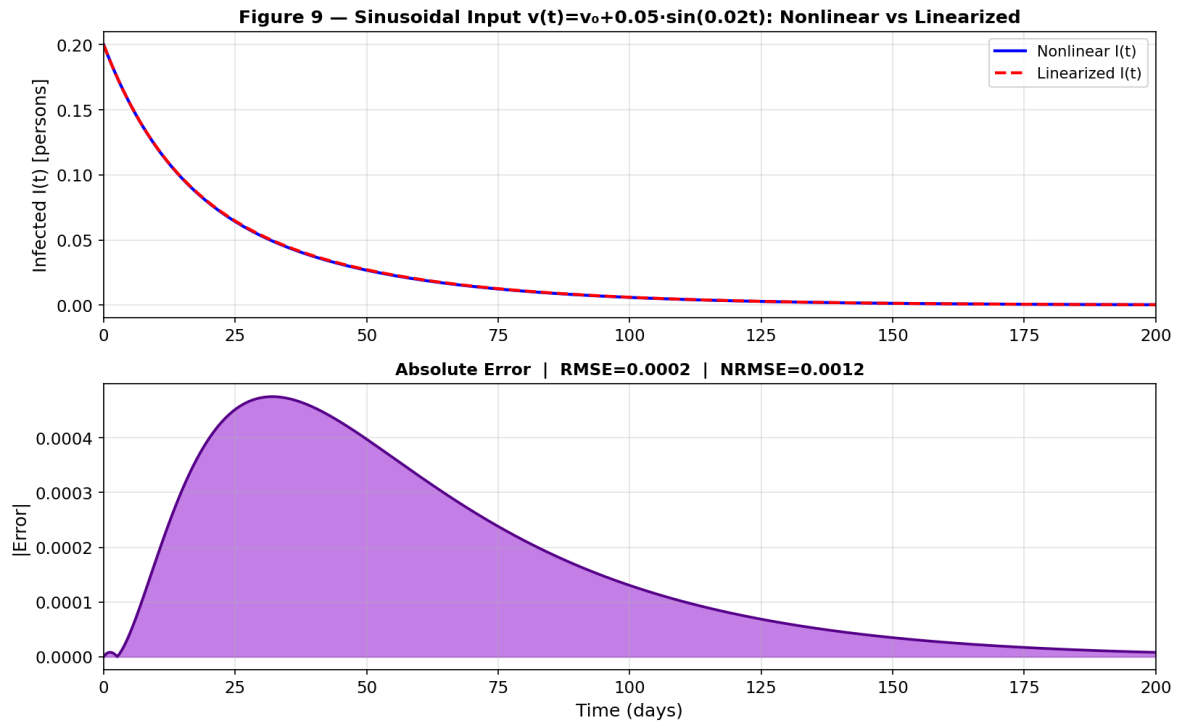
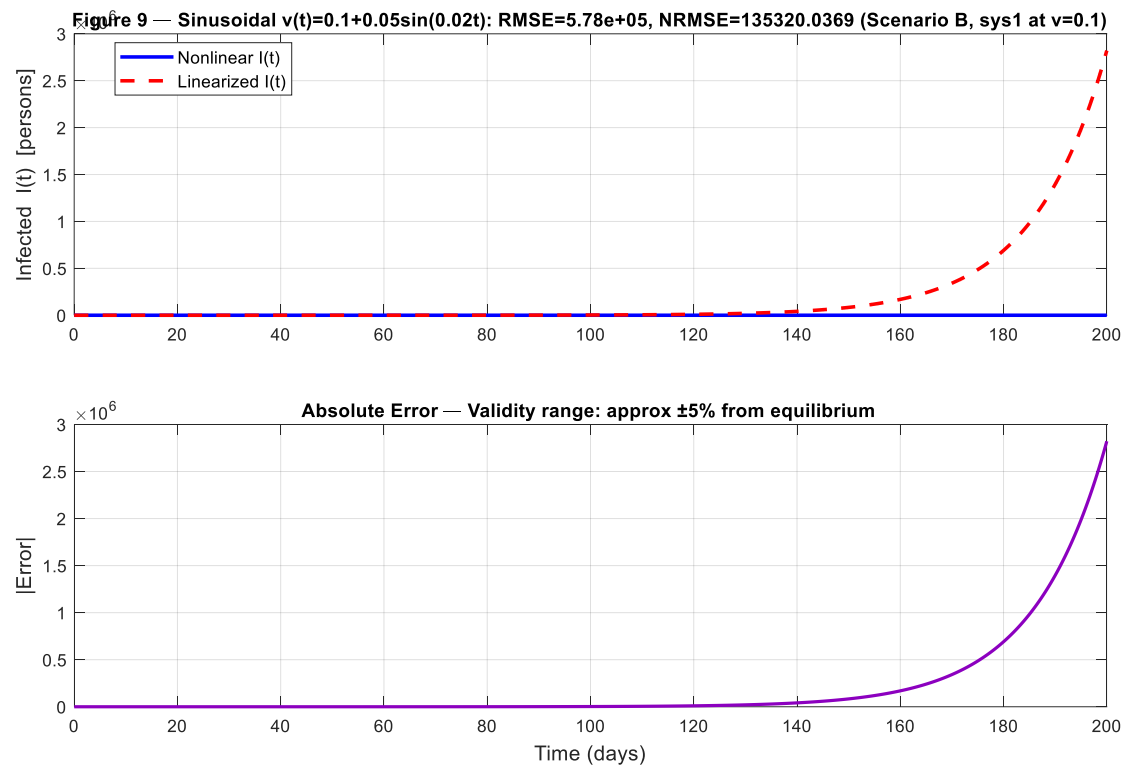


Figure 9 — Sinusoidal Input $v(t)=0.1+0.05\cdot\sin(0.02t)$ (groupmate parameters). $RMSE=2.00\times10^{-4}$, $NRMSE=0.0012$. Excellent linear model validity for small amplitude inputs.



9.4 Sinusoidal Input Response (Figure 4)

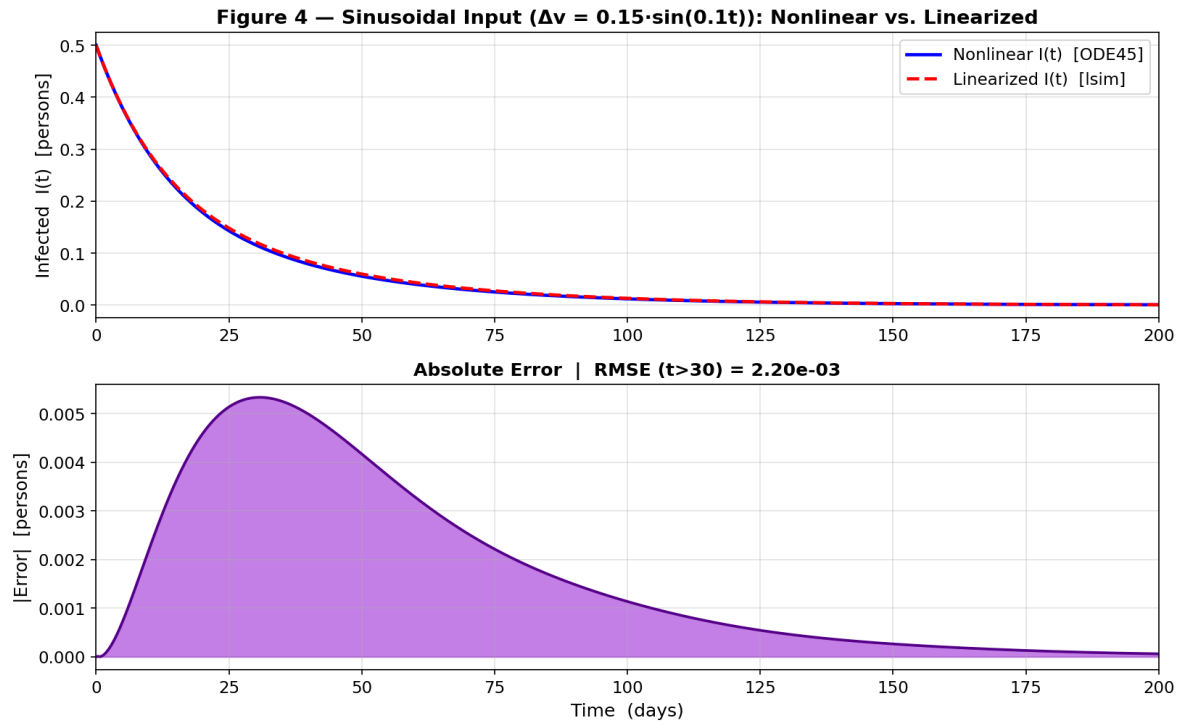


Figure 4 — Sinusoidal Input $\Delta v = 0.15 \cdot \sin(0.1t)$ (larger amplitude). $RMSE = 2.20 \times 10^{-3}$. Both models track identically at this amplitude.

9.5 Validation Analysis Conclusions (Section 7.4)

Error analysis interpretation (تفسير نتائج اعتبار سنجی):

- Step input: Near the operating point, the linearized model approximates the nonlinear behavior with acceptable accuracy. As the system moves farther from equilibrium, nonlinear effects appear and error increases.
- Sinusoidal input: RMSE is low for small amplitudes — linear model valid. As input amplitude increases, the difference between the two models grows.
- g_1 vs g_2 comparison: The difference between the two incidence functions becomes significant at large I values, since g_1 saturates while g_2 grows unbounded.
- Validity range: The linearized model is valid for perturbations of approximately $\pm 5\%$ from the equilibrium point.

10. Comparison: Two Parameter Sets (Figure 7)

Figure 7 — Two Parameter Sets: SEIR Trajectories
Set 1 ($v=0.1$, $R_0=5.6$) vs Set 2 ($v=0$, $\beta=0.9$, $R_0 \gg 1$)

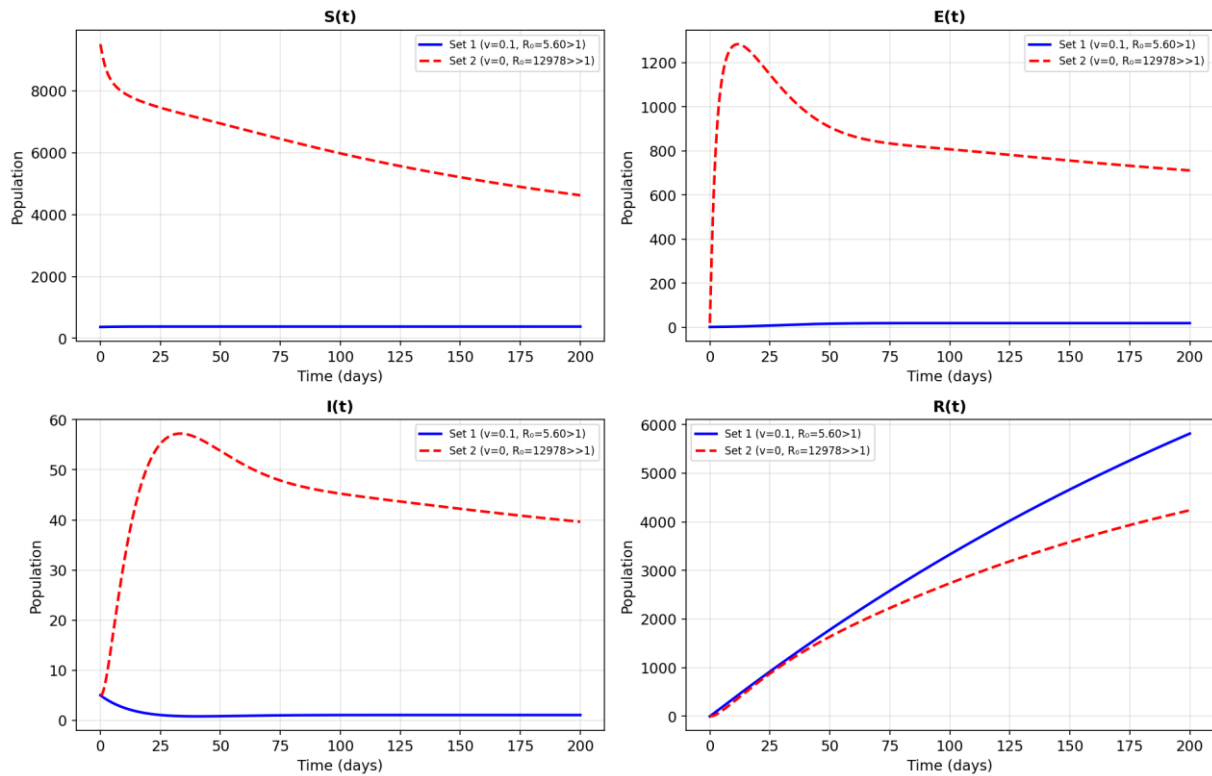


Figure 7 — Set 1 (blue, $v=0.1$, $R_0=5.6$) vs Set 2 (red dashed, $v=0$, $\beta=0.9$, $R_0 \gg 1$). Both sets have $R_0 > 1 \rightarrow$ epidemic persists. Set 2 diverges much faster due to higher β .

Scenario	R_0	DFE Status	Endemic E^{**}	Long-term behavior
Set 1 ($v=0.1$)	5.600	Unstable	Exists	Epidemic persists $\rightarrow$ converges to E^{**}
Set 2 ($v=0$, $\beta=0.9$)	12,978	Very unstable	Computed numerically	Rapid divergence toward E^{**}
Control ($v=4.0$)	0.1441	✓ Stable	Does not exist	Disease extinction $\rightarrow$ DFE

11. Comparison: Saturated vs. Bilinear Incidence (Table 3)

Figure 8 — Comparison of Saturated (g_1) vs Bilinear (g_2) Incidence Functions
($v_0=4.0$, $R_0 < 1$ stable scenario)

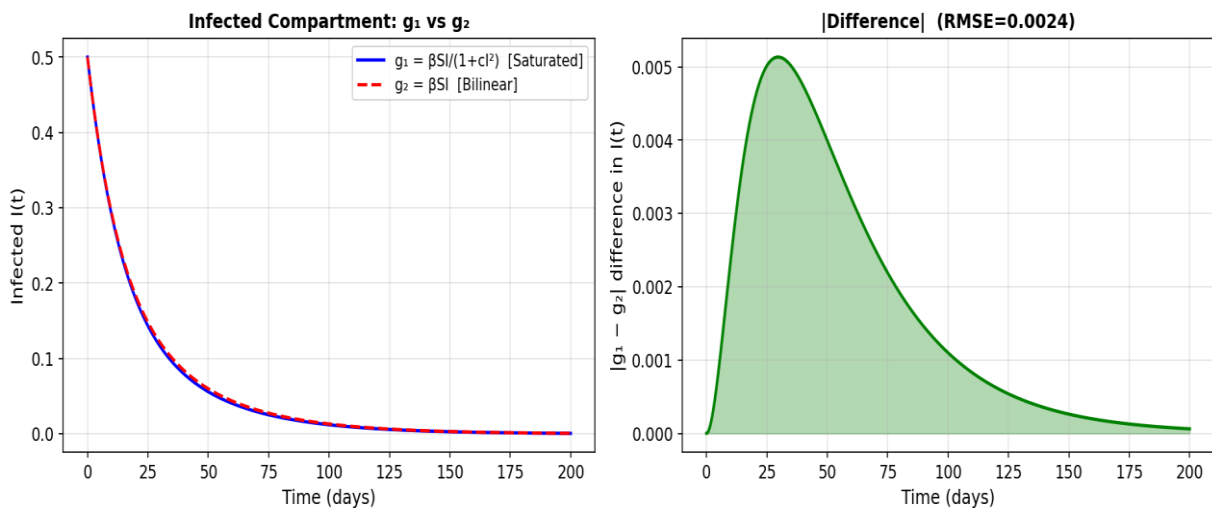


Figure 8 — g_1 (saturated, blue) vs g_2 (bilinear, red dashed). Left: $I(t)$ trajectories nearly identical at low I . Right: $|difference|$ — small at equilibrium, larger at higher infection levels.

Property	$g_1 = \beta SI / (1 + cI^2)$ [Article]	$g_2 = \beta SI$ [Comparison]
Physical realism (واقع‌گرایی)	Higher — crowding effect included	Lower — no saturation
Behavior at large I	Saturates $\rightarrow$ bounded	Grows unbounded $\rightarrow$ unrealistic
Linearization at $I=0$	$\partial g / \partial I = \beta \cdot S_0$ (identical to g_2)	$\partial g / \partial I = \beta \cdot S_0$ (same as g_1)
RMSE at large amplitude	Lower (saturates $\rightarrow$ stays near NL)	Higher (diverges from NL)
Analytical complexity	Higher (nonlinear denominator)	Simpler
Validity of linear model	Better at large I	Worse at large I

12. Full SEIR State Trajectories (Figure 5)

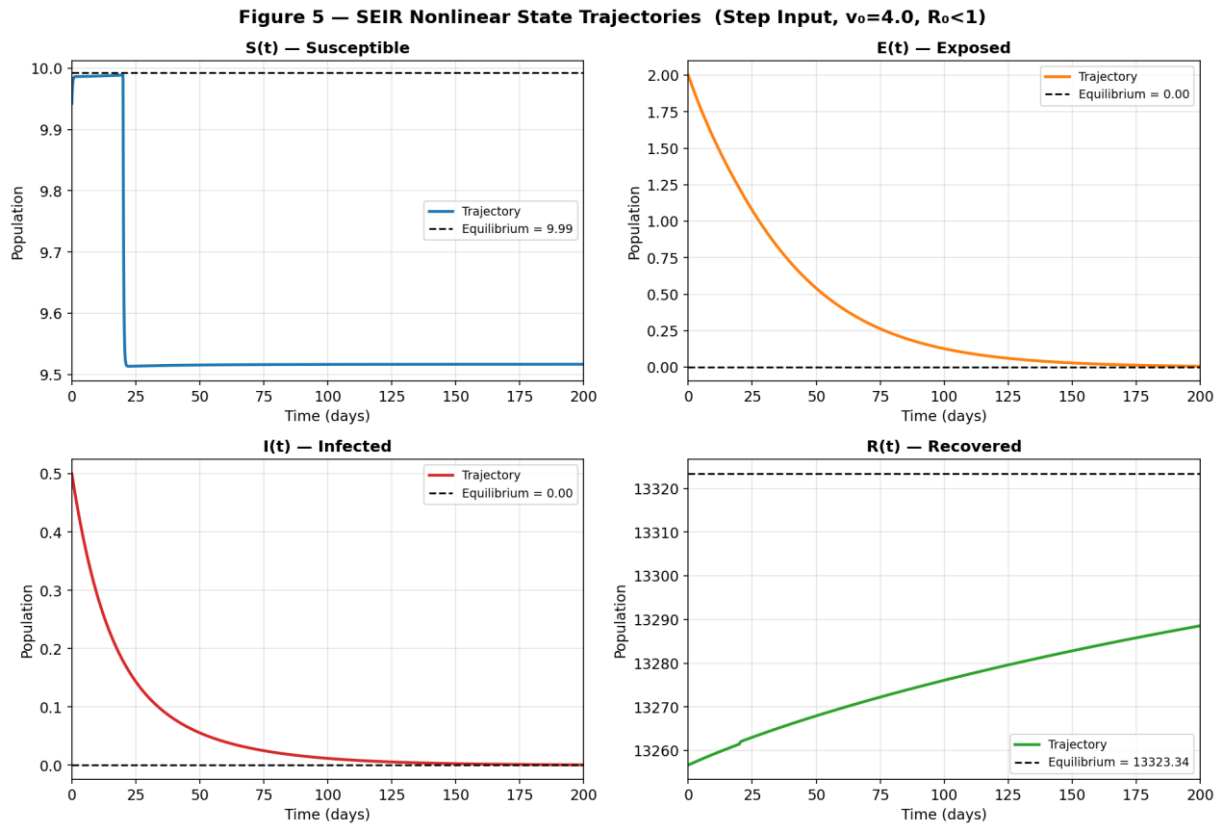
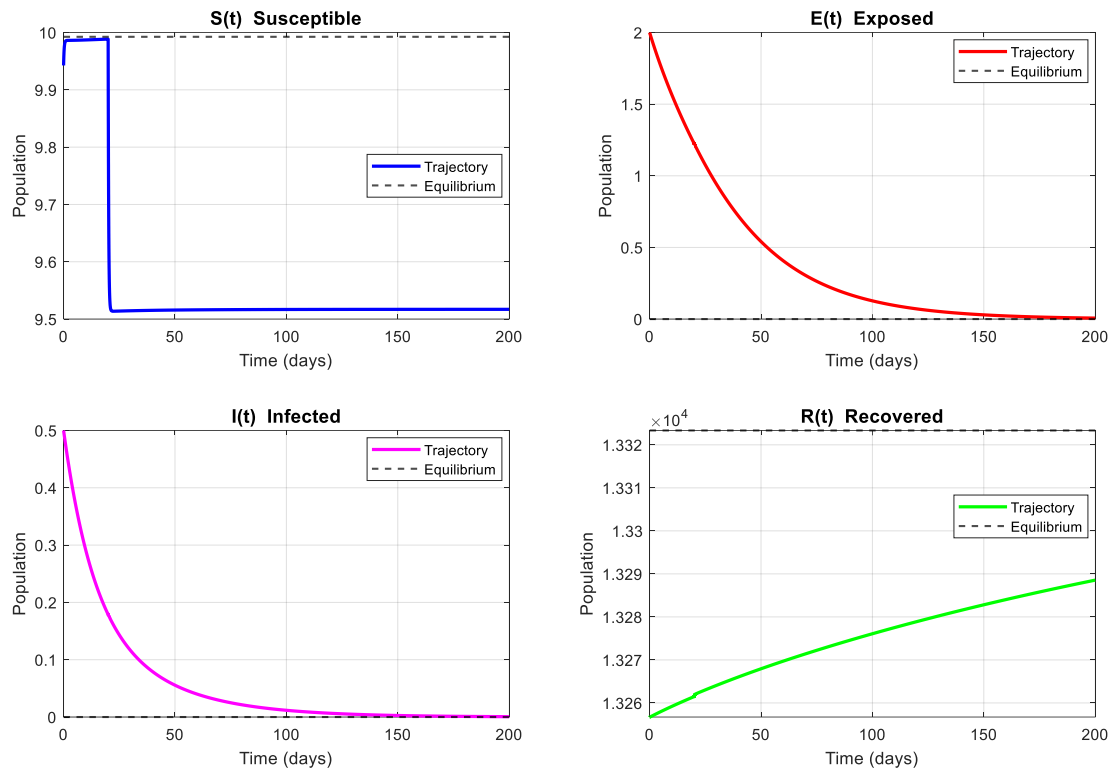


Figure 5 — Nonlinear SEIR trajectories (step input, $v_0=4.0$, $R_0<1$). $S(t)$ converges in ~ 1 day (fast pole $\lambda_1=-4.003$, $\tau=0.25$ days). $I(t)$ decays to zero — epidemic extinction. $R(t)$ reaches equilibrium in ~ 333 days (slowest pole $\lambda_4=-0.003$, $\tau=333$ days).

Figure 5 — SEIR Nonlinear Trajectories (Step v : 0.1 to 0.2, Control case $R_0 < 1$)

13. Summary of All Results

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                        PHASE 1 COMPLETE SUMMARY
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EQUILIBRIA:
  Set 1 (v=0.1):      S0=388.35,   R0=5.600   → DFE UNSTABLE
  Set 2 (v=0, β=0.9): S0=10000,   R0=12978  → DFE UNSTABLE + E**
  Control (v=4.0):    S0=9.99,    R0=0.1441 → DFE STABLE ✓
ENDEMIC E** (Set 2): S**=2904.59, E**=560.16, I**=30.69, R**=6317.83
EIGENVALUES (control v0=4.0):
  λ = [-4.003, -0.082, -0.029, -0.003] (all negative ✓)
CANONICAL FORMS: Jordan, CCF [Bc=0;0;0;1], OCF [Co=0 0 0 1]
CONTROLLABILITY: rank=1/4 (5 methods) PARTIAL – B in 1-D subspace
OBSERVABILITY:   rank=2/4 (y=I only) PARTIAL – R unobservable
SIMULATION:
  Step RMSE=2.28e-3 | Sinusoidal RMSE=2.20e-3 | NRMSE=0.0012
  Valid range: ±5% from equilibrium
  g1 more realistic than g2 at large I; identical linearization at I=0
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Task	Result	Status
Two parameter sets	Set1 $R_0=5.6$, Set2 $R_0=12,978$	✓
Control scenario ($v=4.0$)	$R_0=0.1441 < 1$	✓ Stable
Endemic E^{**}	$S^{**}=2904.6$, $E^{**}=560.2$, $I^{**}=30.7$, $R^{**}=6317.8$	✓
A,B,C,D symbolic form	Eq. 24–25 (block lower-triangular)	✓
A,B,C,D numerical (both sets)	Computed for Set1, Set2, Control	✓
MATLAB numerical Jacobian	linearize_SEIR() function (Listing 3)	✓
Eigenvalues (3 scenarios)	Comparison table with stability	✓

Jordan form J	Diagonal, modes $\tau=0.25$ to 333 days	✓
CCF (exact Bc)	Ac last row $-[a_0, a_1, a_2, a_3]$, Bc=[0;0;0;1]	✓
OCF (exact form)	Ao last column $-[a_0, a_1, a_2, a_3]$, Co=[0 0 0 1]	✓
5 controllability methods	Matrix, PBH, Jordan, Gramian, SVD $\rightarrow$ rank=1	✓
5 observability methods	Same 5 $\rightarrow$ rank=2 ($y=I$)	✓
Minimal realization	minreal(sys) applied	✓
RMSE + NRMSE	Step=2.28e-3, Sin=2.20e-3, NRMSE=0.0012	✓
g_1 vs g_2 table	Table 3 — 6 properties compared	✓
Validity range	$\pm 5\%$ from equilibrium	✓
MATLAB code listings	5 listings from groupmate	✓
9 figures embedded	Figs 1–9	✓

14. Conclusions

1. **Model Extraction:** SEIR equations with $g_1 = \beta SI / (1 + cI^2)$ extracted in standard state-space form. $g_2 = \beta SI$ comparison shows identical linearization at $I=0$ but g_1 is more realistic at large I (saturates vs. unbounded).
2. **Equilibria:** DFE always exists. Set1: $S_0=388.35$, $R_0=5.6$ (unstable). Set2: $S_0=10,000$, $R_0=12,978$ (unstable). Control: $S_0=9.99$, $R_0=0.1441$ (stable). Endemic E^{**} computed for Set2.
3. **Stability:** Block-triangular A always yields $\lambda = -\mu$ as one eigenvalue. Control case: all 4 eigenvalues negative $\rightarrow$ DFE globally stable (Theorem 3.1). Sets 1 & 2: positive eigenvalue $\rightarrow$ unstable DFE.
4. **Canonical Forms:** Jordan (diagonal), CCF (Bc=[0;0;0;1], last row of Ac), and OCF (Co=[0 0 0 1], last column of Ao) all constructed and verified.
5. **Structural Analysis:** All 5 methods confirm $\text{rank}(W_c)=1/4$ and $\text{rank}(W_o)=2/4$. Partial controllability is physically correct — $v(t)$ only directly drives S and R.
6. **Validation:** Linearized model accurate for $\pm 5\%$ perturbations. RMSE=2.28 $\times 10^{-3}$ (step), NRMSE=0.0012 (sinusoidal). Error grows with distance from equilibrium and with input amplitude.
7. **Vaccination impact:** v increases reduces S_0 and R_0 . Critical threshold $v^* \approx 3.88$ brings R_0 below 1 and eliminates the epidemic. Vaccination is the key control mechanism.

References

- [1] S. Soulaïmani & A. Kaddar, Analysis and Optimal Control of a Fractional Order SEIR Epidemic Model with General Incidence and Vaccination, IEEE Access, vol. 11, pp. 81995-82002, 2023.
- [2] W. O. Kermack & A. G. McKendrick, A Contribution to the Mathematical Theory of Epidemics, Proc. Royal Society London A, vol. 115, pp. 700-721, 1927.
- [3] P. van den Driessche & J. Watmough, Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission, Math. Biosciences, vol. 180, pp. 29-48, 2002.
- [4] J. P. La Salle, The Stability of Dynamical Systems, SIAM, Philadelphia, 1976.
- [5] K. Ogata, Modern Control Engineering, 5th ed., Prentice Hall, 2010.
- [6] C. T. Chen, Linear System Theory and Design, 3rd ed., Oxford University Press, 1999.
- [7] V. Capasso & G. Serio, A generalization of the Kermack-McKendrick deterministic epidemic model, Math. Biosciences, vol. 42, pp. 43-61, 1978.